

AI demand drives up electricity supply costs in largest US market to record high

Posted by u/riverdale-74 • • 0 points • 19 comments

The cost of providing electricity in America's largest power market will hit a record high due to soaring demand from artificial intelligence data centres and delays in building new power plants, raising energy prices for consumers.

Grid operator PJM, which covers 13 states and Washington DC, said on Tuesday it procured energy supplies for \$329.17 per megawatt day, a 22 per cent increase compared with the previous year. The organisation will pay power producers \$16.1bn to meet its energy needs from June 2026 to May 2027, a 10 per cent increase compared with the previous year.

The operator said it expects a 1-5 per cent rise for customers in their energy bills, depending on how utilities and states passed on costs.

"It's unpleasant for ratepayers," said Timothy Fox, a managing director at ClearView Energy Partners. "Higher auction prices will result in higher bills for customers."

PJM sets prices at an annual capacity auction where power suppliers bid to provide the region's projected demand.

Earlier this year PJM and some state governments took steps to try to keep power prices lower after last year's capacity auction delivered a \$269.92 per MW-day price — a more than 800 per cent increase from 2023.

In January, under pressure from a complaint that Pennsylvania governor Josh Shapiro filed with the Federal Energy Regulatory Commission, PJM agreed to a "price collar" — a floor and ceiling on prices set at \$175 per MW-day and \$325 per MW-day for the next two years.

However, the latest auction results are a blow to US President Donald Trump's pledge to slash energy prices by 50 per cent. Data published this month by the Labor department showed that electricity prices rose 5.6 per cent over the past year. More broadly, consumer prices increased by 2.7 per cent.

PJM is a regional transmission organisation, which manages the wholesale electricity market and the reliability of the grid. It includes "data centre alley", a region in Virginia which hosts the world's largest concentration of data centres, making it a test case for how the

growth of the sector will affect power prices.

Grid operators and politicians face an escalating dilemma of how to meet booming power demand from the data centres that train and deploy AI, while keeping utility bills from spiralling.

Utilities are spending record sums on building out their generating and transmission infrastructure, some of which is covered directly by big tech companies including Amazon, Microsoft and Meta.

As well as booming demand from data centres, the energy crunch is being fuelled by long interconnection queues and the retiring of ageing power plants. Last year, PJM said 12-30 per cent of its installed capacity could be retired by 2030.

Comments

u/Shawon770 • 1 points • 2025-08-05 06:08 UTC

The AI boom is clearly pushing infrastructure limits. I've been testing out Kumo by SoranoAI, an AI agent that gives you weather and insights with just natural language no code, no API calls, no need to input your location manually. You can even set custom weather alerts just by typing what you want. Tools like this make the power of AI feel way more accessible.

u/Ok_Can_9433 • 1 points • 2025-07-28 13:53 UTC

It's not data centers driving up costs, it's premature retirement of coal plants with no dispatchable generation to replace it.

u/schi854 • 2 points • 2025-07-24 21:29 UTC

Now AI is really begin to eat our lunch. Maybe one day it will take control of the grid to cool itself off before human can get any:). Imagine that happens during a deadly heat wave

u/nspy1011 • 1 points • 2025-07-24 12:22 UTC

AI power needs must be split from consumer and commercial needs and the Microsoft, Meta, Googles of the world need to pay for those (both setting up the infrastructure and operating it). Enough with privatizing profits and socializing costs/losses

u/Puzzleheaded_Ad_3507 • 1 points • 2025-07-24 09:47 UTC

And thanks to greedy billionaires who paid Trump, Americans will be forced to look to China to supply us with the parts for renewable energy

u/Tzilbalba • 1 points • 2025-07-24 03:42 UTC

Goddamn prob why my electrical bill shot up 30% the past 2 months.

u/Future_Helicopter970 • 1 points • 2025-07-24 03:25 UTC

I think this forecast of huge increases in electricity demand will not hold up. As the technology improve electricity demand will become more reasonable. The hardware powering AI is performing faster every year, improving at a faster rate than Moore's Law. These performance gains have been measured by outside groups, so it's not just marketing hype. Specialized AI chips are doing better than general purpose chips.

u/Usual_Retard_6859 • 1 points • 2025-07-26 01:35 UTC

That hasn't happened so far. Dennard Scaling predicted that with smaller transistors per unit area the power consumption would remain constant. This has been proven wrong. Even with lower voltages current draw is increasing. With increased power consumption come increased heat generation and increased costs and power draw of cooling. Global warming will only increase the cooling costs.

Before anyone debates the validity of global warming it's not a question if it's happening, the debate is the cause so please stay on topic.

u/iwerbs • 1 points • 2025-07-28 04:42 UTC

Yeah the cause isn't much of a debate either - even the oil companies' scientists have known since the 1960s that CO2 released from burning coal, gas and oil would eventually trap heat in the Earth's atmosphere. Company brass ordered them to keep quiet about their findings.

u/Lovis1522 • 8 points • 2025-07-23 17:04 UTC

Remember when everyone said it was going to be EV's that did this....

u/Due_Satisfaction2167 • 6 points • 2025-07-23 15:53 UTC

> President Donald Trump's pledge to slash energy prices by 50 per cent.

lol, what sort of idiot actually thought that was going to happen?

Trump promised rate cuts, but all his actual proposed policies demanded rate hikes.

Replacing renewables with more fossil fuels is: 1) Functionally impossible, and 2) Going to cost a fuck of a lot more per MWh.

u/tomrlutong • 2 points • 2025-07-23 15:13 UTC

Don't forget that PJM's gas fleet was downrated 2.8 GW because they're still not reliable in the winter. If not for that, prices would have been around half what they are.

u/Helicase21 • 4 points • 2025-07-23 12:56 UTC

This is a capacity market functioning exactly how it should. Prices are supposed to be high when the capacity position is tight. And the capacity position is in fact tight.

u/HeKnee • 3 points • 2025-07-23 14:05 UTC

But shouldnt all the data centers pay for the increased cost due to increased demand? Utilities are supposed to be regulated to control cost increases.

Why would average consumers be on the hook considering the fact that their usage likely went down due to rooftop solar and energy efficiency improvements?

u/Helicase21 • 1 points • 2025-07-23 14:21 UTC

This is a capacity auction. It's not a direct 1:1 impact on your bill. Each utility has to procure sufficient accredited capacity to meet its load. There certainly are ways to get data centers to be on the hook (for example if a data center signs a power purchase agreement to pay directly for the output of a power plant, but then that output is no longer available to bid in to the broader capacity auction) but data centers aren't the only form of load growth. People buying EVs, replacing gas furnaces with heat pumps, etc all increase electrical load.

u/jabblack • 0 points • 2025-07-23 05:51 UTC

It's an increase of \$65 a year, not a big deal

u/Franklin_Le_Tanklin • 10 points • 2025-07-23 04:11 UTC

Americans are going to get screwed because Trump stopped Biden's tax credits for solar panel factories and wind turbine factories.

Anyone who can't survive off the grid with solar and battery is going to get destroyed in High electricity rates

u/Helicase21 • 2 points • 2025-07-23 13:21 UTC

Even those folks who can go "off grid" still relies on goods and services produced by industries that can't go off grid. So you're still meaningfully exposed to those prices.

u/[deleted] • 1 points • 2025-07-23 11:02 UTC

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